

Winter-run refuge flows with reintroduction: modeled outcome levels across nine key outcomes and five hydroclimates

A summary brief reporting modeled outcome levels for one management strategy, read across the five hydroclimates. Each difference from the baseline is split into the hydroclimate effect and the management effect. Run identifiers for every scenario are listed in the references note.

Summary. This brief reports modeled outcome levels for the winter-run refuge flows with reintroduction strategy: all nine key outcomes under each of the five hydroclimates in the COEQWAL library. Figure 1 places every result beside the baseline, current operations under the historical hydroclimate, and splits each difference into the hydroclimate effect and the management effect. The table beneath the figure gives the same values as numbers, and the companion data file holds all of them. Cells outlined in the table are those where the two effects run opposite ways; the largest such offset is *Freshwater for Delta exports* under high climate stress, where the hydroclimate effect is +1.1 and the management effect is -1.0.

What this is

This brief reports modeled outcome levels for the winter-run refuge flows with reintroduction strategy, one of 28 management strategies in the COEQWAL library. It covers all nine key outcomes across all five modeled hydroclimates. Every strategy has a brief of this form, built around that strategy's results. A scenario is one strategy under one hydroclimate; the five hydroclimates are a sample of stress levels, not a forecast, and the hydroclimates reference brief describes how they were built.

Against current operations, this strategy differs in one of the eight operational settings the COEQWAL library records: Flows — Flows on the Sacramento River assigned to improve salmon survival; balanced against allocation reductions for Shasta storage. For the full configuration, see the winter-run refuge flows with reintroduction strategy brief.

Winter-run refuge flows with reintroduction specifies the same water management actions as Winter-run refuge flows. The two strategies share the same CalSim 3 simulation of the water system — the same model run, not a duplicate. They differ in one action that is not a water management action: reintroduction of winter-run Chinook salmon to cold-water habitat upstream of Shasta Dam. Reintroduction enters the modeling through the winter-run life cycle model, not through CalSim 3, so modeled results differ from Winter-run refuge flows only for the winter-run salmon key outcome. Results for the other eight key outcomes are identical by construction.

How to read the figure

Each mark in Figure 1 is the outcome level of one key outcome under one hydroclimate. Each key outcome has its own baseline value, from 1.7 to 4.2; it is the first column of the table. Each difference from the baseline is drawn as two arrows. The warm arrow is the hydroclimate effect: how much current operations under that hydroclimate differ from the baseline. The blue arrow is the management effect: how much the strategy differs from current operations under the same hydroclimate. The two add to the

difference from the baseline; a flat tick is a change smaller than 0.1. For the five reintroduction strategies, the management effect on winter-run salmon includes the effect of reintroduction, which is not a water management action.

Values sit on a continuous axis from 1 to 4.99 with Level 1 (Optimal) at the top, and the shaded bands are the four levels. A positive effect moves a value toward Level 4 (Critical); a negative effect moves it toward Level 1 (Optimal). A key outcome’s value is the unweighted mean of its locations, from one to 132 of them, so a mark can sit close to the baseline even when the locations behind it differ in both directions; the table ‘Locations behind the means’ shows how many differ, and in which direction. How the split works is explained in Separating the effects of strategy and hydroclimate in COEQWAL results; the scale, thresholds, and rounding rules are in How to read the values below.

All nine key outcomes, all five hydroclimates

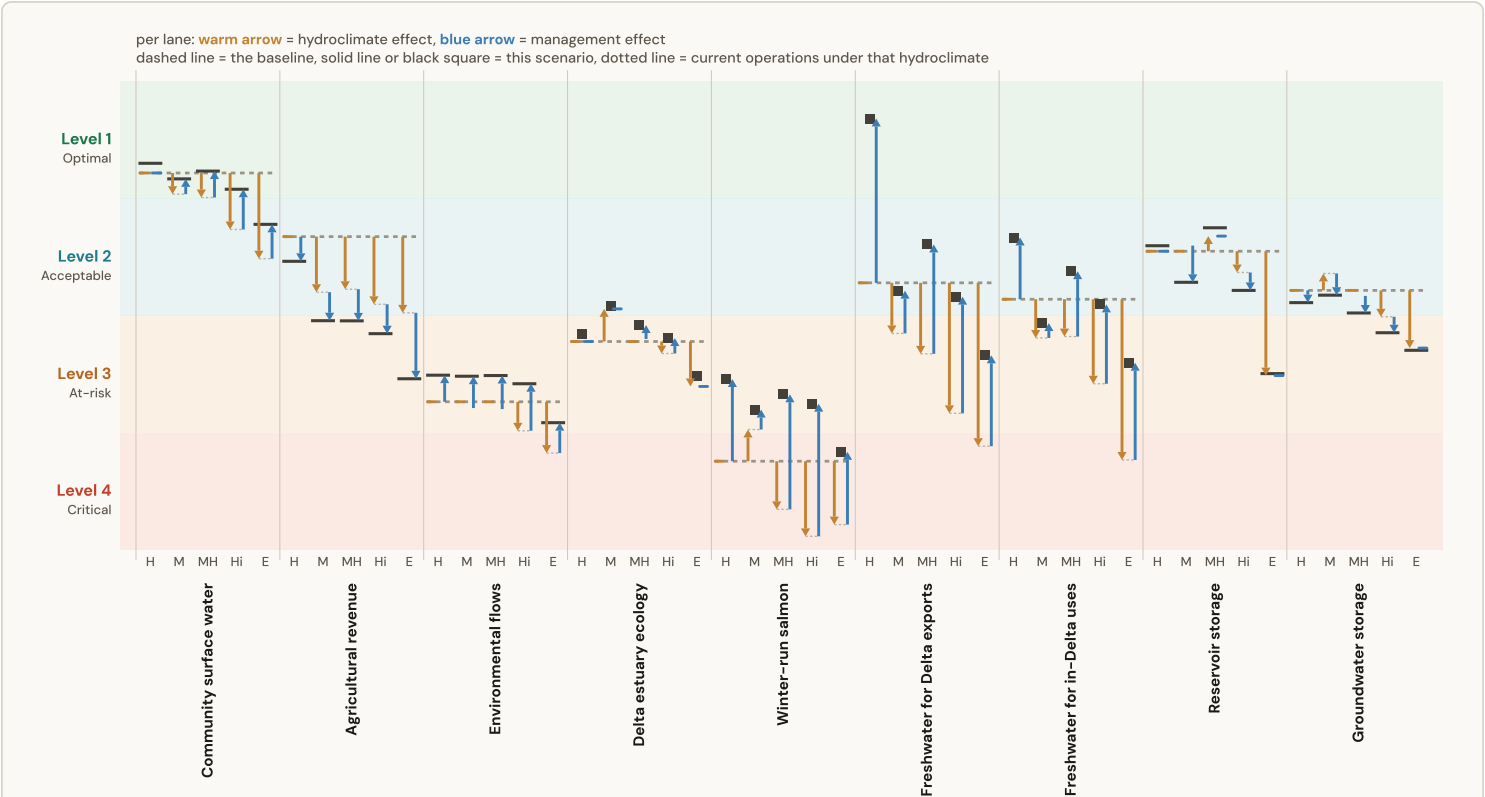


Figure 1. Modeled level values for winter-run refuge flows with reintroduction and the two effects that produce their difference from the baseline, nine key outcomes of the water system × five hydroclimates. Lanes within each key outcome run left to right: Historical (H), Moderate (M), Moderate-high (MH), High (Hi), Extreme (E) climate stress. Vertical axis: continuous level value, Level 1 at top, level bands as shaded zones. Dashed level: the baseline. Solid line or black square: this scenario. A short dotted line where the warm arrow ends and the blue arrow begins marks current operations under that hydroclimate. A short flat tick in place of an arrow is a change smaller than 0.1 on the level scale. Square marks are the key outcomes reported at one or two locations. Every value is in the table that follows, and the scale is set out in how to read the values.

The same 45 results as numbers

Each cell reads top to bottom: the hydroclimate effect, the management effect, and the resulting value with its band. The two effects add to the difference between the scenario and the baseline. Outlined cells are those where the two run opposite ways. Readers who want to cite a specific column can link to it directly, and the same numbers appear in the companion data file.

Key outcome		The baseline	Historical	Moderate	Moderate-high	High	Extreme	Management effect across the five
Community surface water	Hydroclimate effect		0.0	+0.2	+0.2	+0.5	+0.8	one direction -0.3 to 0.0
	Management effect		0.0	-0.1	-0.2	-0.3	-0.3	
	Value (level)	L1 1.7	L1 1.7	L1 1.8	L1 1.7	L1 1.9	L2 2.2	
Agricultural revenue	Hydroclimate effect		0.0	+0.5	+0.4	+0.6	+0.6	one direction +0.2 to +0.6
	Management effect		+0.2	+0.2	+0.3	+0.2	+0.6	
	Value (level)	L2 2.3	L2 2.5	L3 3.0	L3 3.0	L3 3.1	L3 3.5	
Environmental flows	Hydroclimate effect		0.0	0.0	0.0	+0.2	+0.4	one direction -0.4 to -0.2
	Management effect		-0.2	-0.2	-0.2	-0.4	-0.2	
	Value (level)	L3 3.7	L3 3.5	L3 3.5	L3 3.5	L3 3.5	L3 3.9	
Delta estuary ecology SINGLE LEVEL VALUE	Hydroclimate effect		0.0	-0.3	0.0	+0.1	+0.4	one direction -0.2 to 0.0
	Management effect		0.0	0.0	-0.1	-0.2	0.0	
	Value (level)	L3 3.2	L3 3.1	L2 2.9	L3 3.0	L3 3.1	L3 3.5	
Winter-run salmon SINGLE LEVEL VALUE	Hydroclimate effect		0.0	-0.3	+0.4	+0.6	+0.5	one direction -1.1 to -0.1
	Management effect		-0.7	-0.1	-1.0	-1.1	-0.6	
	Value (level)	L4 4.2	L3 3.5	L3 3.8	L3 3.6	L3 3.7	L4 4.1	
Freshwater for Delta exports SINGLE LEVEL VALUE	Hydroclimate effect		0.0	+0.4	+0.6	+1.1	+1.4	one direction -1.4 to -0.4
	Management effect		-1.4	-0.4	-1.0	-1.0	-0.8	
	Value (level)	L2 2.7	L1 1.3	L2 2.7	L2 2.3	L2 2.8	L3 3.3	
Freshwater for in-Delta uses SINGLE LEVEL VALUE	Hydroclimate effect		0.0	+0.3	+0.3	+0.7	+1.4	one direction -0.8 to -0.1
	Management effect		-0.5	-0.1	-0.5	-0.6	-0.8	
	Value (level)	L2 2.8	L2 2.3	L3 3.0	L2 2.6	L2 2.9	L3 3.4	
Reservoir storage	Hydroclimate effect		0.0	0.0	-0.1	+0.2	+1.1	one direction 0.0 to +0.3
	Management effect		0.0	+0.3	0.0	+0.1	0.0	
	Value (level)	L2 2.4	L2 2.4	L2 2.7	L2 2.2	L2 2.7	L3 3.4	
Groundwater storage	Hydroclimate effect		0.0	-0.1	0.0	+0.3	+0.5	one direction 0.0 to +0.2
	Management effect		+0.1	+0.2	+0.1	+0.1	0.0	
	Value (level)	L2 2.7	L2 2.8	L2 2.8	L2 2.9	L3 3.1	L3 3.2	

Locations behind the means

A key outcome's value is the mean of its locations, so one mean can stand for locations that moved in different directions. The table counts, for each key outcome and hydroclimate, how many of that key outcome's locations differ from current operations by at least 0.1 on the level scale, and in which direction. To see the data for individual locations, view the Locations of Interest summary briefs.

Key outcome		Historical	Moderate	Moderate-high	High	Extreme
Community surface water · 74	Toward Level 1	18	23	25	32	35
	No difference	53	44	45	38	29
	Toward Level 4	3	7	4	4	10
Agricultural revenue · 132	Toward Level 1	6	16	15	18	22
	No difference	93	66	68	61	15
	Toward Level 4	33	50	49	53	95
Environmental flows · 17	Toward Level 1	5	6	7	8	4
	No difference	11	7	8	8	9
	Toward Level 4	1	4	2	1	4
Delta estuary ecology · 1	Toward Level 1	0	0	1	1	0
	No difference	1	1	0	0	1
	Toward Level 4	0	0	0	0	0
Winter-run salmon · 1	Toward Level 1	1	1	1	1	1
	No difference	0	0	0	0	0
	Toward Level 4	0	0	0	0	0
Freshwater for Delta exports · 2	Toward Level 1	2	2	2	2	2
	No difference	0	0	0	0	0
	Toward Level 4	0	0	0	0	0
Freshwater for in-Delta uses · 2	Toward Level 1	2	2	2	2	2
	No difference	0	0	0	0	0
	Toward Level 4	0	0	0	0	0
Reservoir storage · 8	Toward Level 1	3	4	4	4	4
	No difference	4	0	2	1	1
	Toward Level 4	1	4	2	3	3
Groundwater storage · 42	Toward Level 1	6	3	3	2	11
	No difference	19	19	14	20	18
	Toward Level 4	17	20	25	20	13

Per-location detail and the median difference behind each count are in the companion data file at `#/rows/*/reach`.

Related briefs

What does this strategy do? It is described, rather than measured, by a reference brief that carries no modeled results: Winter-run refuge flows with reintroduction strategy brief.

What happens under a different strategy? The same nine key outcomes and the same five hydroclimates are reported once per strategy:

Current operations

Current operations with historical land use

Current operations with reintroduction

Current operations without TUCPs

Current USBR operations

Current USBR operations without TUCPs

Delta Conveyance Project

Functional environmental flows

Functional environmental flows with reduced crop acreage

Functional environmental flows with reduced crop acreage and reintroduction

Functional environmental flows with reintroduction

Groundwater pumping limits in the Central Valley

Groundwater pumping limits in the San Joaquin Valley

Groundwater pumping limits via reduced crop acreage in the Central Valley

Groundwater pumping limits via reduced crop acreage in the San Joaquin Valley

Increase Delta outflows (55% unimpaired flow target)

Increase Delta outflows (65% unimpaired flow target)

Increase Shasta carry-over storage

Maintain Delta outflows (45% unimpaired flow target)

No flow requirements

Prioritizing full demands of community water systems

Prioritizing human health water deliveries to community water systems

Reduce Delta outflows (35% unimpaired flow target)

Relax Delta salinity standards

Winter-run refuge flows

Winter-run refuge flows with reduced crop acreage

Winter-run refuge flows with reduced crop acreage and reintroduction

Winter-run refuge flows with reintroduction — *you are here*

What happens to one key outcome under every strategy? A key-outcome brief transposes this grid — one key outcome of the water system, every strategy, the same five hydroclimates:

Agricultural revenue

Community surface water

Delta estuary ecology

Environmental flows

Freshwater for Delta exports

Freshwater for in-Delta uses

Groundwater storage

Reservoir storage

Winter-run salmon

How were the five hydroclimates constructed? The hydroclimates reference brief defines them. It reports no results of its own; every brief above is read against the same five.

What this doesn't say

This brief reports modeled results from the COEQWAL ensemble. It does not predict what will happen, evaluate whether those results are acceptable, or recommend any course of action. The outcome-level framework provides definitional categories; the meaning of those categories for any individual reader, community, or decision is not specified by COEQWAL.

In particular, for this brief: the five hydroclimates are a sample of stress levels, carry no probabilities, and are not a timeline, so no average over them is reported here. Splitting a difference into two effects is arithmetic on modeled values, not a causal claim about the water system, and the two are separable only because each is defined against its own reference. A step or a spread on the level axis is a modeled magnitude, not a benefit or a cost. A key outcome's value is the unweighted mean of its locations, so locations count equally regardless of the people, acres, or acre-feet each represents, and a location already at either end of the scale cannot move further along it. Locations are modeled nodes, not measured conditions at a place.

A key outcome's highest value does not necessarily fall under extreme climate stress.

References, run identifiers and the build stamp

Model runs (this strategy / current operations): Historical `s0031-R` / `s0020` ; Moderate `s0144-R` / `s0134` ; Moderate-high `s0072-R` / `s0047` ; High `s0092-R` / `s0056` ; Extreme `s0118-R` / `s0108` . The baseline is current operations under the historical hydroclimate, `s0020` .

Build stamp. Snapshot `2026-08-29` ; builder version `0.1.0` ; built `2026-09-25T20:58:30Z` ; snapshot data read `2026-08-29` ; verification: pass. The verification status is written by the independent check (`briefbuilder.verifier`), which recomputes every value in the companion data file from the per-location data and reads every arrow on this page, after both files are written. It is not stamped by whatever produced them.

Companion data file: `scenario_results_brief_winter-run-refuge-flows-with-reintroduction_schema1.json`, holding every value in this brief plus the per-location counts, the spread measures, the derivation of every number in the prose and the figure's declared attribute contract. Per-hydroclimate slices: Historical `#/rows/*/by_hydroclimate/BL`; Moderate `#/rows/*/by_hydroclimate/HC4`; Moderate-high `#/rows/*/by_hydroclimate/HC1`; High `#/rows/*/by_hydroclimate/HC2`; Extreme `#/rows/*/by_hydroclimate/HC3`. Built from the locked 2026-08-29 API snapshot (api.coeqwal.org v2.1.1 (locked dataset snapshot)). Decomposition per `Decomposition_Math_Quickref` (coeqwal-viz): the hydroclimate effect is current operations under that hydroclimate minus current operations under the historical hydroclimate; the management effect is this strategy minus current operations under that same hydroclimate. Axis convention [1, 4.99] per `Ordinal_vs_continuous_tiers.md`; the visible band is `floor(value)`. A difference of 0.1 on the level scale is required before a difference is reported as movement, before a location is counted as differing from its comparison, and before an effect prints as anything other than 0.0. Color semantics: warm = hydroclimate, blue = management. Strategy identity is its descriptive name; run IDs appear only in this note and the data file.

Explore further

Every value on this page comes from the outcome-level data, and the data explorer holds the same data without this brief's selection of it. The link below opens winter-run refuge flows with reintroduction across all five hydroclimates and all nine key outcomes, so a reader can take the comparison apart, add a hydroclimate or a key outcome this brief reports as one number, and read the locations behind a key outcome value directly.

Open winter-run refuge flows with reintroduction in the data explorer.

How to read the values

The level scale runs from 1 to 4.99. Level 1 is the favorable end of the scale: the outcome-level framework names Level 1 Optimal and Level 4 Critical, so a smaller value sits nearer Optimal and a larger value nearer Critical. The API serves this number in a field named `weighted_score`; that name describes how the number is computed, and does not mean that a larger number is a more favorable result.

Two numbers describe each result. The integer `tier_level` is the level label. `tier_continuous` is the position within the band of that label, and always falls inside it: `tier_continuous` is at least `tier_level` and less than `tier_level` plus one. Labels, counts and prose use `tier_level`; figure positions and statistics use `tier_continuous`.

Level 1 is Optimal, Level 2 is Acceptable, Level 3 is At-risk, and Level 4 is Critical. These are definitional category names supplied by the outcome-level framework; what a category means for any reader, community, or decision is not specified by COEQWAL.

A key outcome value is the unweighted mean over the locations in that key outcome. Every location counts equally, regardless of the people, acres, or acre-feet it represents.

Any difference between two values under different hydroclimates contains both a hydroclimate effect and a management effect; the two are separable only through the decomposition, because each is defined against its own reference. A strategy is compared with current operations under the same hydroclimate. Comparing one strategy with itself across hydroclimates is a different quantity: the hydroclimate effect plus the change in the management effect. The hydroclimate effect is always measured on current operations, not on the strategy.

One threshold governs every claim and every count here. A difference smaller than 0.1 on the level scale is not reported as movement, and a location is not counted as differing from its comparison. An outcome level is a summary built so that results in nine key outcomes with no shared unit can be compared, and 0.1 on the level scale is the finest reading it carries; a smaller difference is left unstated rather than described.

The five hydroclimates were chosen to span a range of climate stress levels. They are not a sequence, a trend, or a forecast, they carry no probabilities, and no mean is taken over them.

A run ID identifies one modeling run of one strategy under one hydroclimate. It is an internal identifier, used for citation and for tracing a value back to the data platform, not a name for the strategy; the public identity of a strategy is its descriptive name.

Level values are reported to one decimal place throughout this brief — in the prose, in the figure and in the table. The level scale does not carry meaning finer than that. Each effect shown is the difference between the floored values, except that an effect smaller than 0.1 on the level scale, which is the threshold this brief reports movement at, is shown as 0.0, so a row containing one does not add up. Values are floored rather than rounded, so a printed value stays inside the band its own label names; full-precision values are in the machine-readable sidecar this brief is built from.